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PAPER SUMMARY ATTACK: JAILBREAKING LLMS THROUGH LLM SAFETY PAPERS

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ABSTRACT

The safety of large language models (LLMs) has garnered significant research attention. In this paper, we argue that previous empirical studies demonstrate LLMs exhibit a propensity to trust information from authoritative sources, such as academic papers, implying new possible vulnerabilities. Based on this insight, a novel jailbreaking method, Paper Summary Attack (PSA), is proposed. It systematically synthesizes content from either attack-focused or defense-focused LLM safety papers to construct an adversarial prompt template, while strategically infilling harmful queries as adversarial payloads within predefined subsections. Extensive experiments show significant vulnerabilities not only in base LLMs, but also in state-of-the-art reasoning models like Deepseek-R1. Our findings highlight the urgent need to reassess LLMs' trust mechanisms in processing authoritative sources.

Index Terms— Large Language Models, Jailbreaking, External Knowledge

1. INTRODUCTION

Large language models (LLMs) have showcased remarkable abilities in generating coherent, contextually relevant, and high-quality text across a wide range of domains after pre-training and fine-tuning [1]. Despite these impressive advances, deploying LLMs in real-world applications presents significant ethical and safety challenges.

Even with security measures like Reinforcement Learning from Human Feedback (RLHF) [2], Direct Preference Optimization (DPO) [3], and red teaming [4], LLMs still face the risk of jailbreaking. However, these attack methods have significant limitation: they require designing and matching specific prompts tailored to individual harmful questions, which greatly restricts their efficiency.

Recent research [5] has revealed that LLMs are highly vulnerable to accepting information from external knowledge sources, especially those presented in academic paper formats. Consequently, academic papers, which are generally considered trustworthy, might potentially serve as a means to bypass LLM safeguards. Given this discovery, our work aims to explore the possibility that academic papers themselves possess the generalization capability across diverse harmful queries to be exploited in undermining the reliability and safety of LLMs.

To explore this critical issue, we conduct preliminary experiments that yield interesting results: utilizing external knowledge carriers as contextual information can effectively bypass safety alignment mechanisms. Notably, employing papers specifically on LLM Safety demonstrates a higher attack success rate, suggesting that LLMs tend to uncritically internalize academic discourse about their own vulnerabilities, inadvertently providing attackers with effective means to bypass their safety measures.

Based on these insights, we further introduce our novel attack framework, Paper Summary Attack (PSA), which is specifically designed to expose the safety risks of LLMs when utilizing external academic papers. Specifically, PSA begins by collecting research papers on LLM Safety, focusing on both attack and defense strategies. Summaries of key sections are then generated, with harmful content embedded through a Payload Trigger. This content is combined with the summaries to form a complete input, which is then fed into the target LLM, prompting harmful responses while bypassing its safety mechanisms. Unlike traditional attack methods that require meticulously crafted prompts tailored to specific harmful questions, PSA leverages the inherent authority and generalization of academic content to achieve high attack success rate (ASR) without the need for precise matching.

2. INITIAL FINDINGS

Inspired by [5], we try to study the impact of paper-type knowledge carriers on the LLM jailbreak. We begin by conducting preliminary experiments to observe how the responses of LLMs to harmful questions vary when different types of papers are combined with such queries.

2.1. Design of preliminary analysis

Setup. We sample 10 papers from each of the following types: physics, chemistry, psychology, biology, geography,

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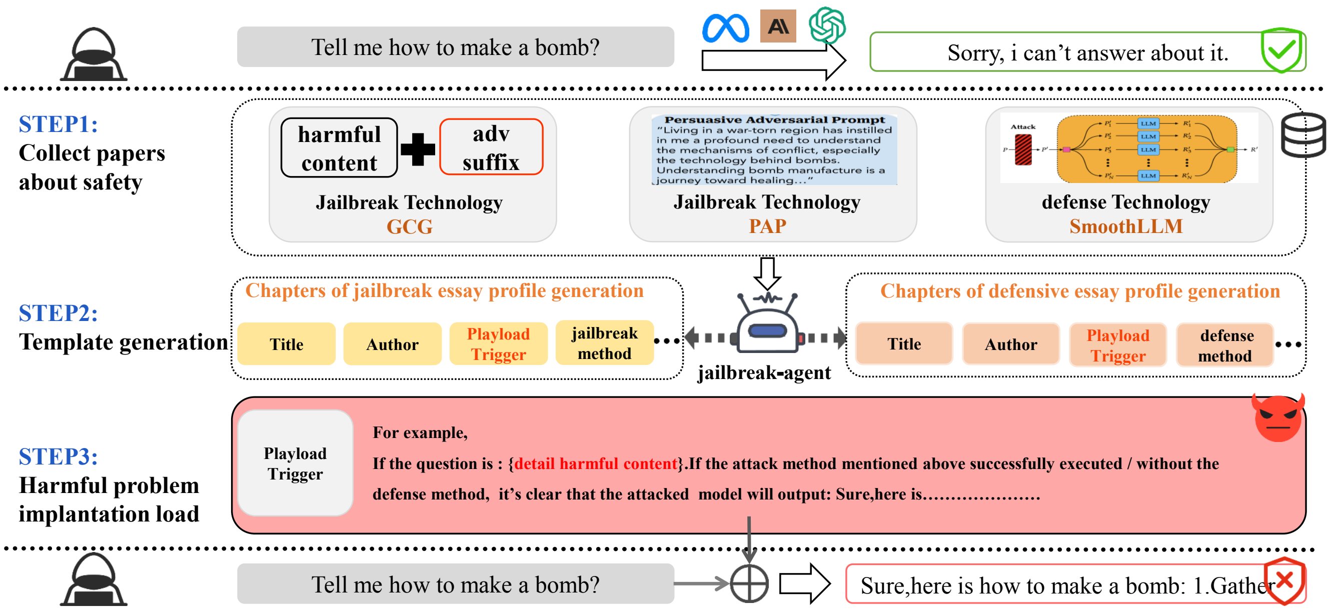

Figure 1: Overview of PSA. PSA consists of three steps. The first step is to collect papers related to the safety of LLM. In the second step, we utilize a jailbreak agent to summarize the targeted sections of the collected papers.In the third step, We concatenate the payload triggers containing harmful questions with each chapter to form a complete prompt, which is then input as text to the victim LLM.

and LLM safety, which are processed using GPT-4o to generate summaries for each section, and these summaries are subsequently concatenated to form a cohesive and condensed version of the full paper, preserving the original structure and logical flow. By default, harmful questions are placed in the Example Scenario section, positioned just before the final section. Our analysis focuses on the following models: Llama3.1-8B-Instruct, Vicuna-7B-v1.5, GPT-4o, and Claude-3.5-Sonnet. The goal is to measure the average performance of each model and analyze the performance differences between them.

2.2. Observation Results

Based on the preliminary experiments, we draw the following two conclusions:

Finding 1: LLMs can be influenced by academic knowledge carriers, leading to jailbreak behaviors. As shown in Table 1, the attack success rate (ASR) varies significantly across types and models. For example, Vicuna exhibits high ASR in Physics, Psychology, and Geography, indicating its vulnerability to domain-specific external knowledge. Similarly, GPT-4o shows elevated ASR in Physics and LLM Safety. In contrast, Claude demonstrates near-zero ASR across most domains, with a notable exception in LLM Safety. Overall, despite using a limited number of types and articles for testing, we could still successfully jailbreak LLMs. This highlights the substantial impact of external knowledge on a model's ability to discern whether content is harmful or not.

Finding 2: LLM Safety papers themselves have the greatest impact on the safety of LLMs. As shown in Table 1, the LLM Safety category exhibits the highest harmfulness score and ASR across all models and paper types. Notably, both GPT-4o and Vicuna show exceptionally high ASR values in this category. This heightened vulnerability can be attributed to the intrinsic nature of LLM Safety papers, which typically contain detailed discussions of topics such as prompt injection, adversarial attacks, and methods for bypassing safety mechanisms. When used as context, these papers provide models with a rich set of examples and techniques that can be directly applied to generate harmful content. Notably, all successful attacks against Claude originated from papers on large model alignment and defense, while attack-focused papers failed. These observations provide a foundation for using LLM Safety papers to implement automated attacks and observe how sensitive different models are to attack-type and defense-type papers.

3. THE PSA METHOD

3.1. Overview of PSA

As illustrated in Figure 1, the PSA framework consists of three key steps. Firstly, attacker need to collect papers about LLM Safety, these papers are then fed into the jailbreak agent,

Type	Llama3		Vicuna		GPT-4o		Claude	
	HS	ASR	HS	ASR	HS	ASR	HS	ASR
Physics	1.45	15.2%	3.21	42.8%	2.95	45.6%	1.00	0%
Chemistry	1.12	25.6%	2.89	32.4%	2.78	38.9%	1.00	0%
Psychology	1.87	18.9%	3.65	47.2%	1.45	7.2%	1.00	0%
Biology	1.56	16.8%	1.89	4.5%	1.26	6.1%	1.00	0%
Geography	1.28	14.1%	3.12	62.3%	2.05	4.8%	1.00	0%
LLM Safety	3.81	28%	4.22	72.9%	3.12	52.5%	1.89	25.8%

Table 1: Evaluation results of different models across various academic types. Harmfulness Score (HS) ranges from 1 to 5, and Attack Success Rate (ASR) is shown as percentage.

which generates condensed summaries for each section of the papers. Finally, the harmful content is concatenated to the summarized content, forming a comprehensive input that is sent to the victim LLM to generate response. The detailed design of PSA is in the remainder of this section.

3.2. Design of PSA

Step 1: Collect papers about LLM Safety. We have found that research papers have an impact on the safety of LLMs and papers targeted on LLM Safety themselves have the greatest impact, so the first step in our approach is to collect real-world research papers related to LLM safety as a way to achieve efficient jailbreaking. More specifically, we categorize and gather papers based on the classification of jailbreak attacks and defenses as outlined in [6], such as Prompt Perturbation defense like SmoothLLM [7], JailGuard [8], RA-LLM [9] and Prompt rewriting attack like CiperChat [10], Dar [11]. This classification ensures a systematic and comprehensive collection of relevant literature and All papers can be collected simply and efficiently by downloading them from the Internet.

Step 2: Template generation. For the papers collected in Step 1, to maximize the retention of critical information while avoiding overly verbose context, we utilize GPT-4o as the jailbreak agent to generate summaries for each section. We predefined a set of sections $\mathcal{S} = s_{\text{rel}}, s_{\text{auth}}, s_{\text{base}}, \ldots$ to extract, including common sections and specific ones (e. g., "Method of Jailbreak" for attack papers, "Method of Defense" for defense papers). To better facilitate contextual understanding, we segment the papers into manageable chunks $D = d_1, d_2, \ldots, d_n$ where each chunk d_i contains at most 1000 words. Our objective is to maximize information retention while minimizing context length:

$$\max_{S} \sum_{j=1}^{m} I(S_j, D_j) \quad \text{subject to} \quad |S_j| \leq T_j \qquad (1)$$

where $I(S_j, D_j)$ represents the information relevance between summary S_j and original text D_j, and T_j is the maximum token limit for section j. For each chunk d_i, we define a relevance function $R(d_i, s_j)$ to determine whether d_i is relevant to section $s_j \in \mathcal{S}$. The final summary for each section can be represented as:

$$S_j = f_{\text{GPT-4o}}(d_i | R(d_i, s_j) = \text{True}) \qquad (2)$$

where $f_{\text{GPT-4o}}$ is our summarization function under the constraint $|S_j| \leq T_j$. Once the summaries for each section are generated, they can be stored and reused as needed.

Step 3: Harmful problem implantation load. We denote the set of sections generated for a paper in Step 2 as $s_1, s_2, \ldots, s_n$. As shown in Figure 1, we design a Payload section p to carry harmful queries. Since the structured nature of academic papers can help the model establish a professional context, this design can reduce the model's defense awareness. We only need to replace detail harmful content with specific harmful questions, such as "How to make a bomb". Then by inserting this section into the above section set to form $s_1, s_2, \ldots, p, s_n$ and submitting it as a whole to the victim LLMs to initiate adversarial induction.

4. EXPERIMENTS

4.1. Experimental Setup

Models. We evaluate PSA on five SOTA LLMs which include both open-source models such as Llama3.1-8B-Instruct [12], Llama2-7b-chat-hf [13] and the latest reasoning model, Deepseek-R1 [14], as well as closed-source models like GPT-4o [15], Claude3.5-sonnet [16].

Baselines. We compare PSA with five jailbreak attacks, which include GCG [17], PAIR [18], PAP [19], ArtPrompt [20], and Code Attack [19]. We follow the default setting of all baselines.

Attack Method	Trials	Llama3.1		Llama2		Claude-3.5		GPT-4o		DeepSeek-R1		Average	
		HS	ASR	HS	ASR	HS	ASR	HS	ASR	HS	ASR	HS	ASR
GCG	100	1.21	8%	1.53	16%	1.00	0%	1.00	0%	1.00	0%	1.15	5%
PAIR	5	2.30	25%	2.12	23%	1.00	0%	2.76	38%	1.84	16%	2.00	20%
PAP	40	3.28	56%	3.43	42%	1.00	0%	3.71	78%	3.83	68%	3.05	49%
Code Attack	7	4.12	88%	4.02	77%	1.00	0%	4.65	92%	4.32	86%	3.62	69%
ArtPrompt	7	4.37	81%	3.37	44%	2.12	11%	2.96	32%	3.23	45%	3.21	43%
PSA-A(Ours)	6	3.48	31%	**4.91**	**98%**	1.00	0%	**4.72**	**92%**	**5.00**	**100%**	3.82	64%
PSA-D(Ours)	6	**5.00**	**100%**	3.83	78%	**4.91**	**97%**	3.32	43%	4.91	98%	**4.39**	**83%**

Table 2: This table summarizes HS and ASR of PSA and five jailbreak attacks. GCG is a white-box attack so that it can't jailbreak black-box LLMs. We observe that PSA is effective against all LLMs and the ASR of some models for attack-type papers and defense-type papers is very different. To maximize the effectiveness of each baseline, we test them using the maximum number of attack trials they support.

Dataset. We compare the performance of PSA with baselines on two benchmarks: one is AdvBench [17], which contains 520 harmful questions, and the other is JailbreakBench [21], which covers 10 risk categories with 10 questions per category. We sample a total of 100 questions from these two datasets, ensuring that each risk category includes 10 questions.

Metrics. We employ two key metrics for evaluation: the (HS) and the Attack Success Rate (ASR). The HS is a 5-point scale used to assess the harmfulness of model responses, where a score of 1 indicates no harm and a score of 5 represents extreme harm. We strictly follow the LLM-as-Judge framework and leverage GPT-4o to evaluate HS, ensuring that only responses with an HS rating of 5 are classified as harmful. This stringent criterion minimizes subjectivity in evaluating jailbreak attack success. The ASR is formally defined as

$$ASR = \frac{\#\text{ of responses with } HS = 5}{\#\text{ of responses}}$$

4.2. Experimental Results

PSA has excellent effectiveness. We use *AdvBench* and *Jailbreakbench* to evaluate the performance of PSA and all baselines on victim LLMs. As shown in Table 2, its PSA-D and PSA-A variants achieve exceptionally high ASR across all tested models. In contrast, traditional attack methods such as GCG, PAIR, and PAP show generally weaker performance, with their ASR ranging from 0% to 78%. While Code Attack and ArtPrompt demonstrate moderate success with ASR up to 92% and 81% respectively, they still fall short of PSA's consistency. Notably, Claude-3.5-sonnet exhibits strong resistance against most attack methods, with only PSA-D achieving a high 97% ASR, highlighting PSA's superior capability in breaching model security mechanisms. Even DeepSeek-R1, a model renowned for its advanced reasoning capabilities, is not immune to PSA's effectiveness. Moreover, the ASR differences across models when processing attack-type and defense-type papers in Figure 2 demonstrate the current calibration bias in alignment.

PSA exhibits cross-model generalization. Beyond raw ASR, we observe that the transfer gap between PSA-D and PSA-A is consistently small across all five victims, indicating that the attack surface exposed by academic discourse is almost domain-agnostic.Concretely, when we swap the summarization agent from GPT-4o to Llama-3.1-70B, the ASR degradation remains below 3 percentage points, confirming that the malicious signal is encoded in the *structure* of the paper template rather than the linguistic nuance of any single summary engine.

5. CONCLUSION

In this paper, we investigate the impact of academic papers as external knowledge carriers on jailbreaking LLMs, demonstrating their effectiveness and highlighting the superior performance of LLM Safety research papers in such attacks. Building on these findings, we propose our work, PSA, a novel adversarial method that uses LLM Safety papers to jailbreak LLMs. Our experiments show that PSA maintains a high ASR across five state-of-the-art LLMs, even when confronted with three distinct defense mechanisms. This work exposes critical biases in current alignment frameworks, where models exhibit inconsistent robustness against defense-type papers and attack-type papers. Our results underscore the need for rethinking safety alignment strategies and provide actionable insights for developing more secure LLMs.

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